

Multi-organ/auto-segmentation of MICA CT scans

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Abstract

This study explores the use of Artificial Intelligence (AI), specifically Convolutional Neural Networks (CNNs), to automate the 3D segmentation of multiple organs in non-human primate (NHP) CT scans. The dataset, MICA, consists of CT scans from six NHPs, annotated for 11 organs. Two CNN models, DeepEdit and Full-CT, are utilized for organ segmentation. While DeepEdit, trained on a limited dataset of non-human primates (NHP), shows variable accuracy, Full-CT, pre-trained on human CT scans, surpasses DeepEdit in organs sharing common labels between the MICA dataset and the Full-CT model.

1 Introduction

This report aims to explore the potential applications of Artificial Intelligence (AI), specifically Convolutional Neural Networks (CNNs), in automating the 3D segmentation of multiple organs within computerized tomography (CT) scans related to non-human primates (NHP).

2 Data

The study utilizes CT scans from six NHPs, captured at three distinct time points: days 1, 3, and 7. An expert has annotated all 18 CT scans. Each scan includes annotations for 11 organs: brain, lungs, left ventricle of the heart, heart, small intestine, kidneys, liver, spleen, large intestine, right inguinal lymph node, and bone marrow (refer to Fig. 1). The annotations were conducted using VivoQuant¹ software.

The dataset analyzed in this report, MICA, is a subset of a larger dataset, as presented in Table 1. Not all portions of the dataset were available during the current analysis. Upon its completion, we intend to revisit the analysis for the entire dataset. In addition, IDs associated with 6 NHPs from MICA dataset analyzed here is recorded in Table 2.

Dataset name	Description	Number of			
		NHPs	Timepoints	Labels	CT scans
MICA	anti-MICA mAb injected	6	3	11	18
Sotrovimab	Sotrovimab injected	6	5	?	30
CLDN3 ADC	anti-CLDN3 mAb injected	9	5	?	45

Table 1: Details of CT datasets with corresponding descriptions and counts of Non-Human Primates (NHP), timepoints, organ labels, and available CT scans.

NHP ID (MICA)
A19620
A20334
A20343
A20349
A20351
A21014

Table 2: IDs associated with NHPs in MICA datasets analyzed in this work.

¹<https://www.vivoquant.com/>

For every organ, Fig. 2 displays the distribution of volume measurements across all CT scans. For instance, it illustrates that the lung volumes exhibit the highest maximum measurements, while the right inguinal lymph node volumes occupy the lower end of the volume spectrum.

Figure 1: A 3D view of a sample NHP CT scan (MICA) with annotated organs and their labels.

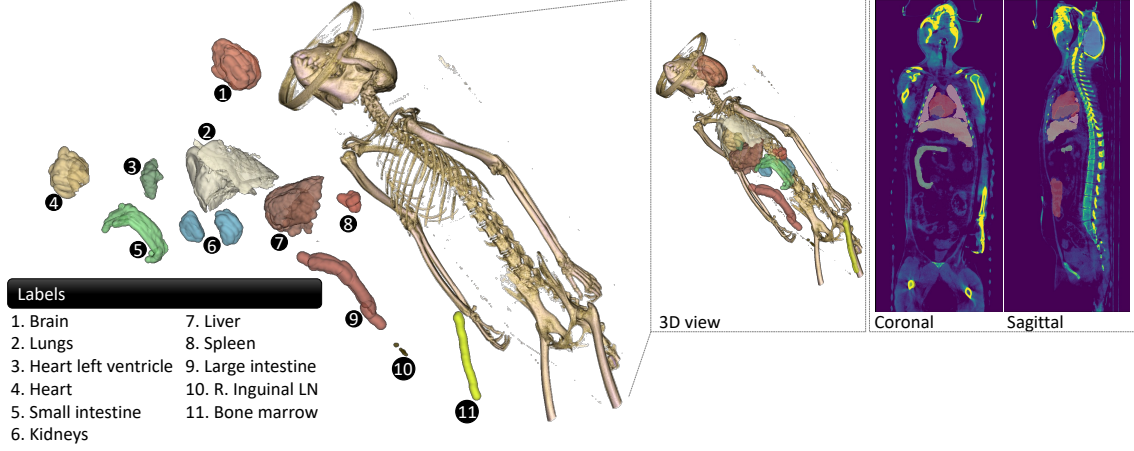
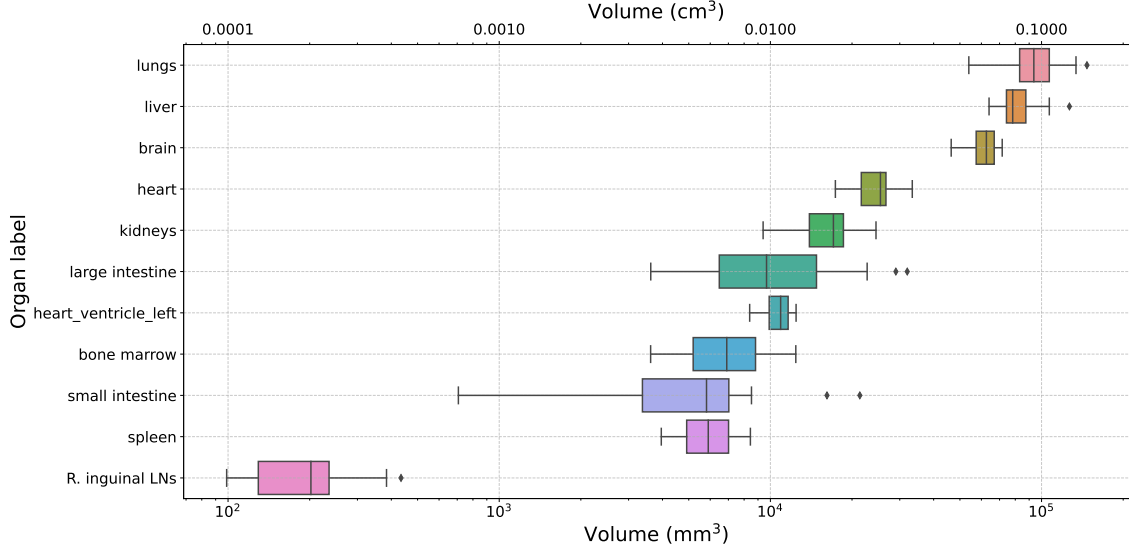


Figure 2: Box plot illustrating volume measurements across all 18 CT scans for each organ, from MICA dataset (see Table 1), in units of mm^3 on the bottom x-axis and cm^3 on the top x-axis. These volume estimations are derived from expert annotations using Slicer software.



3 Analysis

To accomplish the segmentation of organs in CT scans, we utilize MONAI Label², an efficient open-source tool designed for image labeling and AI model training [DPAI⁺22]. Notably, MONAI Label seamlessly integrates with the 3D Slicer platform³ [FBKC⁺12], a widely used open-source software recognized for its excellence in medical image analysis and visualization.

²<https://github.com/Project-MONAI/MONAILabel>

³<https://www.slicer.org/>

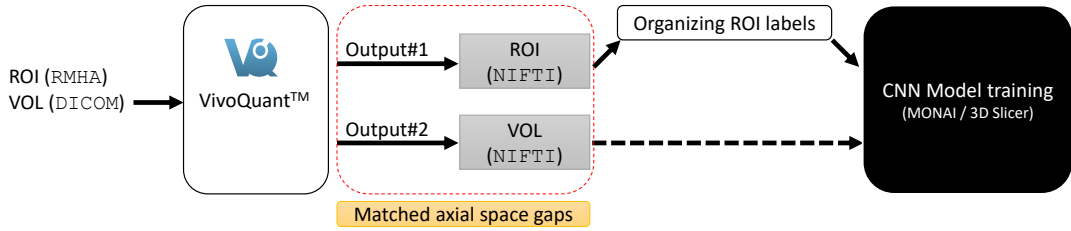
3.1 Preprocessing

In order to train a model using MONAI, the data volumes and their corresponding annotations, such as segmentation masks, must adhere to standard formats like NIFTI⁴, which are compatible with various open-source software. However, the annotations for CT scans performed in VivoQuant have been exported in the proprietary RMHA format⁵. These RMHA files encompass a header containing image metadata and annotated label names, along with a binary section storing the coordinates of the annotated pixels. Consequently, to facilitate compatibility with MONAI, it is necessary to utilize VivoQuant for the conversion of the annotated masks to NIFTI format.

Notably, the original CT scans are in DICOM format and have a 1.0mm spacing along the axial view (Spacing: (0.64, 0.64, 1)). Conversely, the exported segmentation masks from VivoQuant have a different spacing of 0.64mm (Spacing: (0.64, 0.64, 0.64)). To ensure consistency, we also export the CT scans from VivoQuant as NIFTI files. This process ensures that both the segmentation masks and the CT scans have a uniform axial view spacing of 0.64mm (refer to Fig. 3).

As a final procedural step, it is essential to systematically organize the annotated label names, ensuring consistency across various forms such as *Kidney*, *Kidneys*, and potential misspellings like *Kedney*. This harmonization extends to the corresponding label values present in the original RMHA files. To achieve uniformity, these variations must be reassigned and revalued, aligning them with a reference table. This pre-processing step is crucial in preparation for the subsequent stage of model training.

Figure 3: Data preprocessing steps for model training include converting proprietary RMHA format annotations from VivoQuant to standard NIFTI format, ensuring compatibility with MONAI and 3D Slicer. Original DICOM CT scans, initially having 1.0mm axial spacing, are exported alongside segmentation masks with 0.64mm spacing. Both scans (VOL) and masks (ROI) are uniformly converted to NIFTI format. Additionally, systematic organization of annotated label names and their assigned values is performed before model training.



3.2 CNN model training

The initial step in estimating vital parameters from CT scans of any specific organ involves accurate segmentation of that organ. CNNs are highly effective for this purpose. In this study, we utilized two CNN models available in MONAI: DeepEdit[DPMA+22] and Wholebody CT segmentation (Full-CT)⁶. All model training and validations were conducted within a Microsoft Azure Ubuntu virtual environment, leveraging an NVIDIA Tesla V100 GPU.

For each CNN model, we executed multiple training sessions. During each session, we utilized 5 out of 6 available NHP CT scans for training and validation, reserving the remaining scan for testing. This process was iterated through cross-validation, ensuring that each NHP CT scan was used as a test set, not encountered by the model during training or validation.

3.2.1 DeepEdit

The DeepEdit model was trained on all 11 available labels for 300 epochs during each iteration. The correlations between manually estimated volumes and DeepEdit’s predictions on the test dataset are shown in Figure 4. The overall mean accuracy was around 56%, with some labels achieving

⁴<https://nifti.nimh.nih.gov/>

⁵<https://www.vivoquant.com/files/manual/2022/VQ-Manual.pdf>

⁶<https://monai.io/model-zoo.html>

higher values and others achieving lower values. Thus, as observed from the linear regression analysis, some labels are better identified than others. For example, lungs ($R^2 = 0.66$) and liver ($R^2 = 0.7$), which have the largest physical volumes among the organs, are reasonably segmented. However, the segmentation quality is poor for the other organs.

A closer look at the estimated correlations reveals that the prediction is extremely poor when:

- An organ has a small physical volume (e.g., spleen or right inguinal lymph nodes), as estimated from manual annotation.
- An organ has a wide range of values for the limited amount of training samples. For example, the maximum volumes of the large/small intestines measured manually are more than 30/8 times their minimum volumes in our sample of 6 NHPs. Given that only 5 NHPs are used during training/validation, it becomes difficult for a model to converge with the right parameters to correctly predict the volume on an unseen sample.
- There is an inconsistency in segmentation masks: An example within the current dataset is bone marrow, which for a given NHP (e.g., A19620), could be located on either side of the body (left or right) in the segmentation masks.

These may explain why the model sometime predicts zero volumes for organs such as the right inguinal lymph nodes, spleen, bone marrow, and small/large intestines.

3.2.2 Full-CT

The Full-CT model utilizes a pre-trained SegResNet model for volumetric (3D) segmentation across 104 whole-body segments, as detailed by Myronenko et al. [Myr18]. This model was initially trained on the TotalSegmentator⁷ dataset [Was23], which, in turn, leveraged nnUNet⁸ [Ise21] on 1228 human CT scans sourced from diverse scanners, institutions, and protocols.

Of the 104 labels available in the Full-CT pre-trained model, only four align with the MICA dataset, specifically spleen, liver, left ventricle of the heart, and brain. Notably, some organs like intestines are absent from the Full-CT model, while others such as lungs, kidneys, and the heart are subdivided into multiple regions rather than represented as single labels. For example, lungs consist of distinct segments, including `lung_upper_lobe_left`, `lung_lower_lobe_left`, `lung_upper_lobe_right`, `lung_middle_lobe_right`, and `lung_lower_lobe_right`.

Despite the limitations of not covering all labels in our dataset, applying the Full-CT model to the common four labels (spleen, liver, left ventricle of the heart, and brain) allows us to evaluate the impact of using a model extensively trained on human CT scans and its potential to enhance our model training performance.

We conduct Full-CT model training for 100 epochs. In Figure 5, we visualize model predictions against manual measurements for the aforementioned four organs. Notably, unlike the DeepEdit model, the Full-CT model does not generate zero values for predicted volumes. We also observe a robust correlation for the brain ($R^2 = 0.77$), in contrast to a significantly lower value of 0.16 when compared to DeepEdit.

In Figure 6, we present a comparison of the volume ratios derived from DeepEdit and Full-CT predictions against manual measurements. Leveraging normal distribution functions and examining key fitting parameters such as mean (μ) and variance (σ^2), a clear trend emerges – Full-CT consistently outperforms DeepEdit across all organs. This observation underscores the superior training quality of the Full-CT model, despite its training duration being fewer in epochs.

4 Future Work

Our intention is to expand the current analysis to include all studies listed in Table 1 once data collection for each is completed. Insights drawn from the analyses presented in this work highlight several key considerations:

- ****Diversity in Training Data****: Given the diverse array of organs and their volumes, a crucial need for additional training data arises. It is important for a model to encounter various scenarios related to a given organ to effectively learn parameters and accurately predict outcomes. As an example, the Full-CT model is trained on 1228 human CT scans whereas DeepEdit is trained on only 5 NHP CT data.

⁷<https://github.com/wasserth/TotalSegmentator>

⁸<https://github.com/MIC-DKFZ/nnUNet>

- ****Transfer Learning Advantage:**** Our findings underscore the superiority of models trained on human data compared to those trained from scratch. However, to maximize the benefits, maintaining consistency in annotated organ labels between both data and pretrained model input labels is essential.
- ****Active Learning Recommendation:**** Looking ahead, we propose exploring an active learning approach instead of annotating each CT scan from scratch. This strategy facilitates the annotation process, enabling experts to efficiently annotate larger volumes of data.

Additionally, it’s crucial to be cautious when drawing conclusions about model accuracy, especially when comparing outputs with manual measurements. For instance, the data in this study was annotated by a single expert, and the potential inter-observer variability remains unknown. To address this, conducting annotation by multiple experts for a subset of two or three datasets can provide insights into standard errors among different expert annotations. This helps to better understand of how closely a model’s predicted values align with ground truth (GT) values.

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Figure 4: Scatter plots with overlaid linear regression models displaying the comparison between manual (in mm^3) and DeepEdit-derived (in mm^3) volumetric measurements for anatomical features, including the spleen, liver, left ventricle of the heart, brain, heart, kidneys, small intestine, large intestine, bone marrow, right inguinal lymph nodes, and lungs. Each subplot portrays the relationship between the manual and DeepEdit-generated volumes, emphasizing the degree of agreement through linear regression lines (dashed red) and their associated coefficient of determination (R^2) values. Data points are color coded according to NHP IDs.

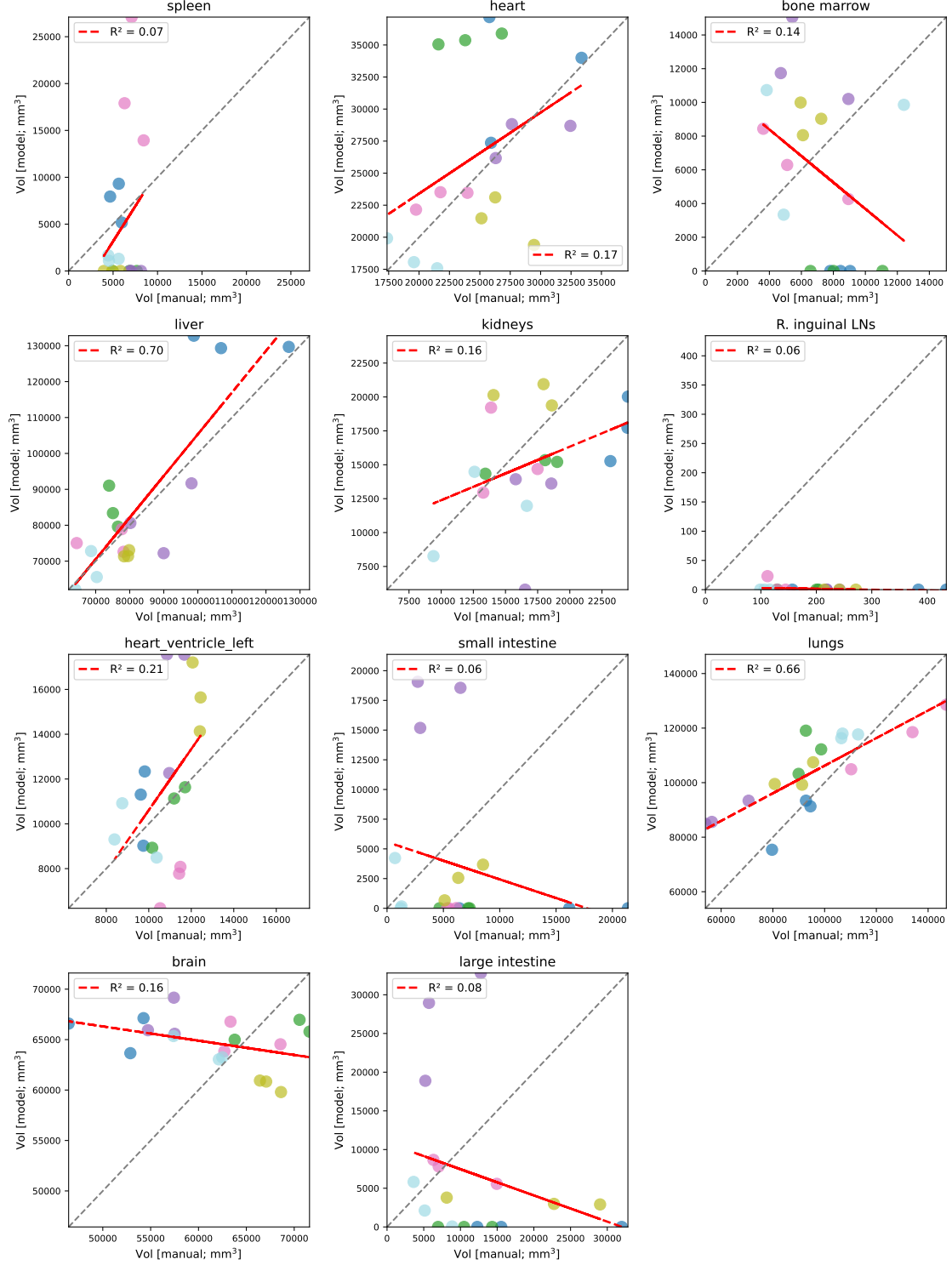


Figure 5: Comparative scatter plots between manually estimated and Full-CT model-derived volumes (cm^3) of anatomical organs including the spleen, liver, left ventricle of the heart, and brain. Linear regression models (dashed red lines) show the correlation between the manual and model-derived volumes, with the coefficient of correlation (R^2) provided for each subplot. Data points are color coded according to NHP IDs.

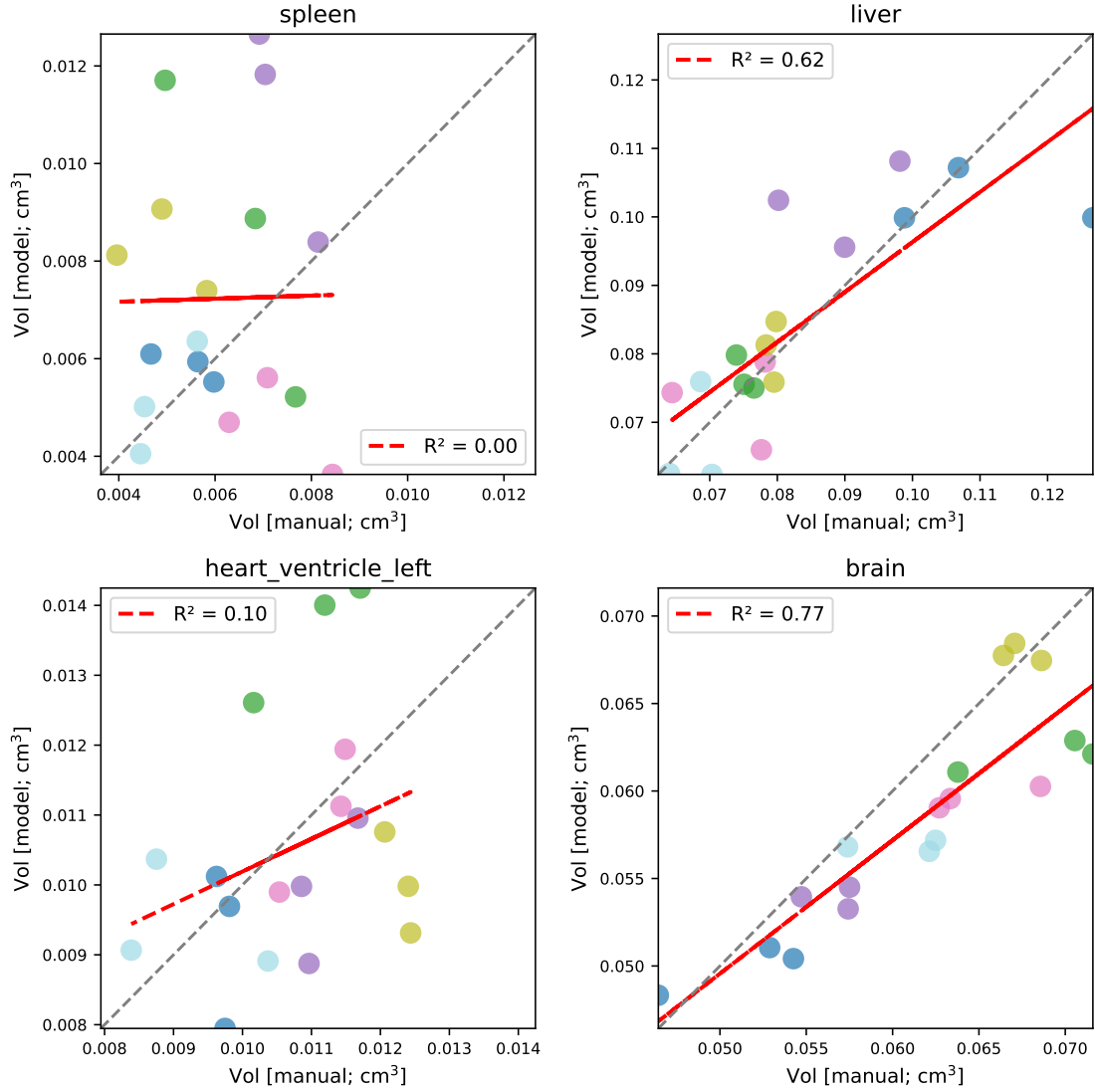


Figure 6: Comparative histograms presenting the distribution of ratios between model-derived and manually obtained volumes across common anatomical features: spleen, liver, left ventricle of the heart, and brain. Each subplot displays histograms representing the ratio of model-derived to manually measured volumes, as obtained from DeepEdit (in lime) and Full-CT (in magenta) models. The dashed grey line indicates a ratio of 1.0, emphasizing parity between the model and manual measurements. Overlaid normal distribution fits (in green for DeepEdit and dark magenta for Full-CT) provide mean (μ) and variance (σ^2) statistics for the respective distributions, illustrating the deviation and distribution characteristics of the ratios.

